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Question 1 [C02] [10 Points]

Design a **BowlingRecord** class where each object stores a Bangladesh player's name and wickets taken in two innings of the first Test match against Australia. Use appropriate variables to track the player's information, record status, and Bangladesh's total wickets. The constructor initializes the player information, and each record is initially not added. The **addRecord()** method adds the player's wickets to the team total only once, while **removeRecord()** removes them.

Tester Code	Output
<pre> public class BowlingRecordTester { public static void main(String[] args) { BowlingRecord.showTeamWickets(); BowlingRecord.showBowlerCount(); System.out.println("1====="); BowlingRecord a = new BowlingRecord("Hasan Mahmud", new int[]{5,4}); a.details(); System.out.println("2====="); a.addRecord(); System.out.println("3====="); BowlingRecord b = new BowlingRecord("Mehidy Hasan Miraz", new int[]{2,4}); b.details(); BowlingRecord.showTeamWickets(); System.out.println("4====="); b.addRecord(); BowlingRecord.showTeamWickets(); System.out.println("5====="); BowlingRecord c = new BowlingRecord("Taskin Ahmed", new int[] {3, 1}); c.addRecord(); c.details(); b.details(); System.out.println("6====="); BowlingRecord d = new BowlingRecord("Nahid Rana", new int[] {1,1}); d.addRecord(); BowlingRecord.showTeamWickets(); System.out.println("7====="); b.removeRecord(); BowlingRecord.showTeamWickets(); System.out.println("8====="); BowlingRecord.showBowlerCount(); c.addRecord(); System.out.println("9====="); c.removeRecord(); c.details(); System.out.println("10====="); BowlingRecord.showTeamWickets(); BowlingRecord.showBowlerCount(); } } </pre>	<pre> Bangladesh Wickets: 0 Total Bowlers: 0 1===== Bowler ID: 1, Name: Hasan Mahmud Innings Wickets: [5, 4] Total Wickets: 9, Recorded: false 2===== Bowler 1 record added. 3===== Bowler ID: 2, Name: Mehidy Hasan Miraz Innings Wickets: [2, 4] Total Wickets: 6, Recorded: false Bangladesh Wickets: 9 4===== Bowler 2 record added. Bangladesh Wickets: 15 5===== Bowler 3 record added. Bowler ID: 3, Name: Taskin Ahmed Innings Wickets: [3, 1] Total Wickets: 4, Recorded: true Bowler ID: 2, Name: Mehidy Hasan Miraz Innings Wickets: [2, 4] Total Wickets: 6, Recorded: true 6===== Bowler 4 record added. Bangladesh Wickets: 21 7===== Bowler 2 record removed. Bangladesh Wickets: 15 8===== Total Bowlers: 4 Bowler 3 record is already added. 9===== Bowler 3 record removed. Bowler ID: 3, Name: Taskin Ahmed Innings Wickets: [3, 1] Total Wickets: 4, Recorded: false 10===== Bangladesh Wickets: 11 Total Bowlers: 4 </pre>

Name:	ID:	Section:
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Question 1 [C02] [10 Points]

Design a **BattingRecord** class where each object stores a Bangladesh player's name and runs scored in two innings of the first Test match against Australia. Use appropriate variables to track the player's information, record status, and Bangladesh's total runs. The constructor initializes the player information, and each record is initially not added. The **addRecord()** method adds the player's runs to the team total only once, while **removeRecord()** removes them.

Tester Code	Expected Output
<pre> public class BattingRecordTester { public static void main(String[] args) { BattingRecord.showTeamRuns(); BattingRecord.showBatterCount(); System.out.println("1====="); BattingRecord a = new BattingRecord("Mominul Haque", new int[] {42, 31}); a.details(); System.out.println("2====="); a.addRecord(); System.out.println("3====="); BattingRecord b = new BattingRecord("Najmul Hossain Shanto", new int[] {61, 24}); b.details(); BattingRecord.showTeamRuns(); System.out.println("4====="); b.addRecord(); BattingRecord.showTeamRuns(); System.out.println("5====="); BattingRecord c = new BattingRecord("Mehidy Hasan Miraz", new int[] {65,18}); c.addRecord(); c.details(); b.details(); System.out.println("6====="); BattingRecord d = new BattingRecord("Litton Das", new int[] {37, 12}); d.addRecord(); BattingRecord.showTeamRuns(); System.out.println("7====="); b.removeRecord(); BattingRecord.showTeamRuns(); System.out.println("8====="); BattingRecord.showBatterCount(); c.addRecord(); System.out.println("9====="); c.removeRecord(); c.details(); System.out.println("10====="); BattingRecord.showTeamRuns(); BattingRecord.showBatterCount(); } } </pre>	<pre> Bangladesh Runs: 0 Total Batters: 0 1===== Batter ID: 1, Name: Mominul Haque Innings Runs: [42, 31] Total Runs: 73, Recorded: false 2===== Batter 1 record added. 3===== Batter ID: 2, Name: Najmul Hossain Shanto Innings Runs: [61, 24] Total Runs: 85, Recorded: false Bangladesh Runs: 73 4===== Batter 2 record added. Bangladesh Runs: 158 5===== Batter 3 record added. Batter ID: 3, Name: Mehidy Hasan Miraz Innings Runs: [65, 18] Total Runs: 83, Recorded: true Batter ID: 2, Name: Najmul Hossain Shanto Innings Runs: [61, 24] Total Runs: 85, Recorded: true 6===== Batter 4 record added. Bangladesh Runs: 290 7===== Batter 2 record removed. Bangladesh Runs: 205 8===== Total Batters: 4 Batter 3 record is already added. 9===== Batter 3 record removed. Batter ID: 3, Name: Mehidy Hasan Miraz Innings Runs: [65, 18] Total Runs: 83, Recorded: false 10===== Bangladesh Runs: 122 Total Batters: 4 </pre>

1.1 Set A Tentative Solution

```
import java.util.*;

public class BowlingRecord {
    static int teamWickets=0;
    static int bowlerCount=0;
    String name;
    int [] wickets;
    boolean recorded=false;
    int id;

    public BowlingRecord(String name,int [] wickets){
        this.name=name;
        this.wickets=wickets;
        bowlerCount+=1;
        id=bowlerCount;
    }

    public static void showTeamWickets(){
        System.out.println("Bangladesh Wickets: "+teamWickets);
    }

    public static void showBowlerCount(){
        System.out.println("Total Bowlers: "+bowlerCount);
    }

    public void details(){
        System.out.println("Bowler ID: "+id+", Name: "+name+"\nInnings Wickets: "+ Arrays.toString(wickets)+
            "\nTotal Wickets: "+(wickets[0]+wickets[1])+", Recorded: "+recorded);
    }

    public void addRecord(){
        if (!recorded){
            System.out.println("Bowler "+id+" record added");
            recorded=true;
            teamWickets+=(wickets[0]+wickets[1]);
        }
        else
            System.out.println("Bowler " +id+ " record is already added.");
    }

    public void removeRecord(){
        System.out.println("Bowler " +id+ " record removed.");
        teamWickets--=(wickets[0]+wickets[1]);
        recorded=false;
    }
}
```

1.2 Rubric for Set A

SI	Points to Meet	Marks (10)
1.	Proper Java syntax and structure	1
2.	Correct constructor implementation and ID assignment	2
3.	Properly defines and uses static variables and methods	2
4.	<code>details()</code> displays all required information correctly	2
5.	<code>addRecord()</code> and <code>removeRecord()</code> correctly update wickets and record status	3
Total		10

Set B Tentative Solution(s)

```
import java.util.*;
public class BattingRecord {
    static int teamRuns=0;
    static int batterCount=0;
    String name;
    int [] runs;
    boolean recorded=false;
    int id;

    public BattingRecord(String name,int [] runs){
        this.name=name;
        this.runs=runs;
        batterCount+=1;
        id=batterCount;
    }
    public static void showTeamRuns(){
        System.out.println("Bangladesh Runs: "+teamRuns);
    }
    public static void showBatterCount(){
        System.out.println("Total Batters: "+batterCount);
    }
    public void details(){
        System.out.println("Batter ID: "+id+", Name: "+name+"\nInnings Runs: "+
Arrays.toString(runs)+
        "\nTotal Runs: "+(runs[0]+runs[1])+", Recorded: "+recorded);
    }
    public void addRecord(){
        if (!recorded){
            System.out.println("Batter "+id+" record added");
            recorded=true;
            teamRuns+=(runs[0]+runs[1]);
        }
        else
            System.out.println("Batter " +id+ " record is already added.");
    }
    public void removeRecord(){
        System.out.println("Batter " +id+ " record removed.");
        teamRuns-=(runs[0]+runs[1]);
        recorded=false;
    }
}
```

Rubric for Set B (Coding)

SI	Points to Meet	Marks (10)
1.	Proper Java syntax and structure	1
2	Correct constructor implementation and ID assignment	2
3	Properly defines and uses static variables and methods	2
4	<code>details()</code> displays all required information correctly	2
5	<code>addRecord()</code> and <code>removeRecord()</code> correctly update wickets and record status	3
Total		10